CRP 4680/5680: Introduction to Urban Data Science – Spatial Regression II: Geographically weighted regression

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City and Regional Planning
Spring 2025

Course Plan

Section 0: Python environment setup and jupyter lab (2 classes)

Section 1: Introduction to Python and Data Techniques (11 classes)

- Python basics; Data management; Data visualization; Geospatial operations; Data collection

Section 2: Exploratory Spatial Data Analysis (ESDA) and Spatial Econometrics

- Spatial weights and spatial dependence (3 classes)
- Regression and spatial regressions (4 classes)

Section 3: Machine Learning (5 classes)

- Unsupervised learning methods
- Supervised learning methods
- Guest Speaker: Prof.Wanshan Qiu (from HKU) – April 17 (waiting for his response)

Updated Syllabus

- Please see the new syllabus at: Canvas -> Modules ->The latest version of the syllabus

- The final project presentation cancelled. ***The 5% originally allocated for the presentation will be given to everyone as a bonus***.

- Aside from this, ***there are no other changes to the syllabus***. The grading criteria will be followed as outlined in the original syllabus.

HW 4 is due on April 20

Part1: Spatial autoregressive regressions

Part2: GWR analysis

Part3: Initial Methods Exploration of the Final Project

In-class exercise for Week 10 and this week are due on this Friday

Final Project Guideline

- Format:
  - ***A report/research paper*** using the applied techniques of urban data science and GIS that emphasizes ***spatial*** analysis and ***spatial-related*** questions
  - A separate ***html file*** documents the details of data processing, modeling, and visualization steps in your project (For explanatory purposes only; grading will be based on your final report.).
- Minimum requirement:
  - ***Individual work***: double-spaced, min 12 pages excluding references, 12 font
  - ***Group work:*** double-spaced, min 17 pages excluding references, 12 font
  - ***Graduate student requirement (CRP 5680):*** you are required to prepare a literature review in final report. Your review should summarize the key contributions of previous studies, discuss their implications, and analyze how they relate to your research question.
- Expectations:
  - ***The project should be analytical, not just descriptive*** – use at least one analytical methods we covered (e.g., spatial correlation, spatial regressions, machine learning methods), or other analytical methods you learned from other related courses (e.g., network analysis, suitability analysis). Simply generating data through spatial operations (e.g., joins, merges) and visualizing isn't enough.
  - Use this final project to assist your exit projects and client-based tasks.

Final Project Guideline

Structures of the report/research paper:

- 1. Intro – Research Background, literature review, research questions, intro to the research region (20 pts)
- 2. Research Methods – Describing the methods and your steps (roadmap if necessary) (15 pts)
- 3. Data & description - Data source, data cleaning and preparation (summary statistics, data transformation, variable construction, justification for classification method etc.) (15 pts)
- 4. Spatial analysis results & the explanation and interpretation of results. (25 pts)
- 5. Conclusion, implications, and limitations (10 pts)
- 6. Appendix: reference etc. (5 pts)

The rest 10 pts: Complete Structure, right formatting, meets minimum page requirement, Clear and organized writing

Final Project Guideline

- Submission: May 16 at 11:59pm via Canvas
  - Final research report/paper – in pdf format.
  - Html file: exported from your Jupyter Notebook.

Recap – spatial regression

- OLS regression

- Spatial regression models that capture spatial dependence
  - spatial lag model (SLM)
  - spatial error model (SEM)
  - spatially lagged X model (SLX)

- After spring Break: Spatial regression models that capture spatial heterogeneity
  - Geographically weighted regression (GWR)

Geographically Weighted Regression (GWR)

- Geographically Weighted Regression (GWR) allows relationships in a regression model to vary over space
- Local estimation based on nearby location
- Yields a different coefficient for each location

Spatial heterogeneity refers to the idea that relationships between variables or patterns in data can vary across different locations or geographic areas.

ASU SPARC Spatial Analysis Research Center

Arizona State University

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Message from the director

The goal of scientific research is typically to answer the question, *"Why are things the way they are?"*

We are often prompted to ask this question by observing that data are not randomly distributed in space but exhibit clustered patterns of highs and lows. One of the goals of SPARC is to try to uncover information on the *processes* that have generated these data. Why did some counties vote Republican in the last US Presidential election and why did some vote Democrat? Why are rates of heart disease so much higher in the south-eastern states of the US compared to the rest of the country? Why are crime rates higher in some parts of a city than in others?

These questions are not easy to answer because they involve spatial dependency in data, possible spatial heterogeneity in processes, and, almost certainly, dependency on spatial scale. However, SPARC has an internationally renowned reputation for answering these questions with expertise in geographic information science, computational spatial science, spatial statistical analysis and spatial visualization.

Please feel free to contact us with any spatial problems you might have.

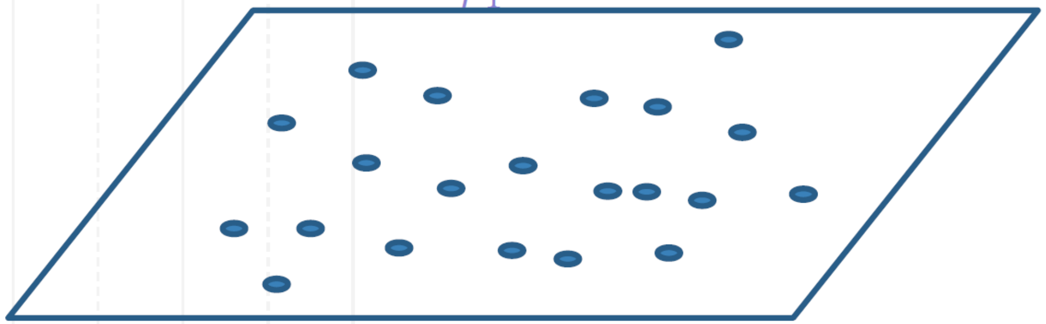
– A. Stewart Fotheringham, Director of SPARC

Geographically weighted regression

- **Global Regression**

$$y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$$

Same β_0 for each one
Same β_1 for each one

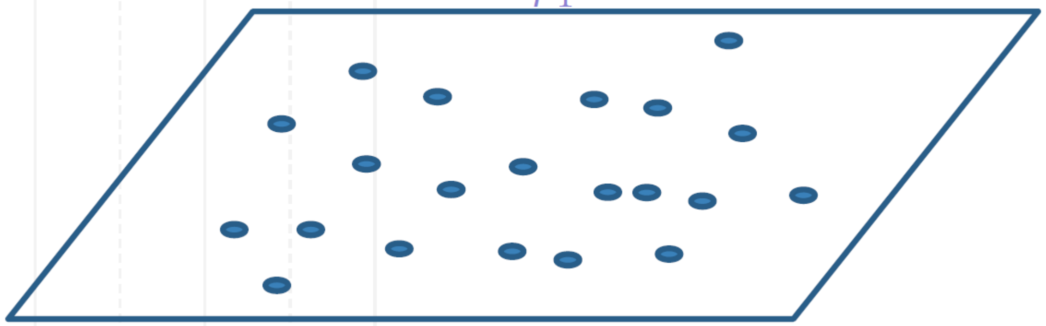


- **Local Regression**

$$y_i = \beta_{0i} + \beta_{1i} x_{i1} + \varepsilon_i$$

Varying over i

Different β_0 for each one
Different β_1 for each one



GWR

$$y_i = \beta_0(u_i, v_i) + \beta_1(u_i, v_i)x_{i1} + \varepsilon_i$$

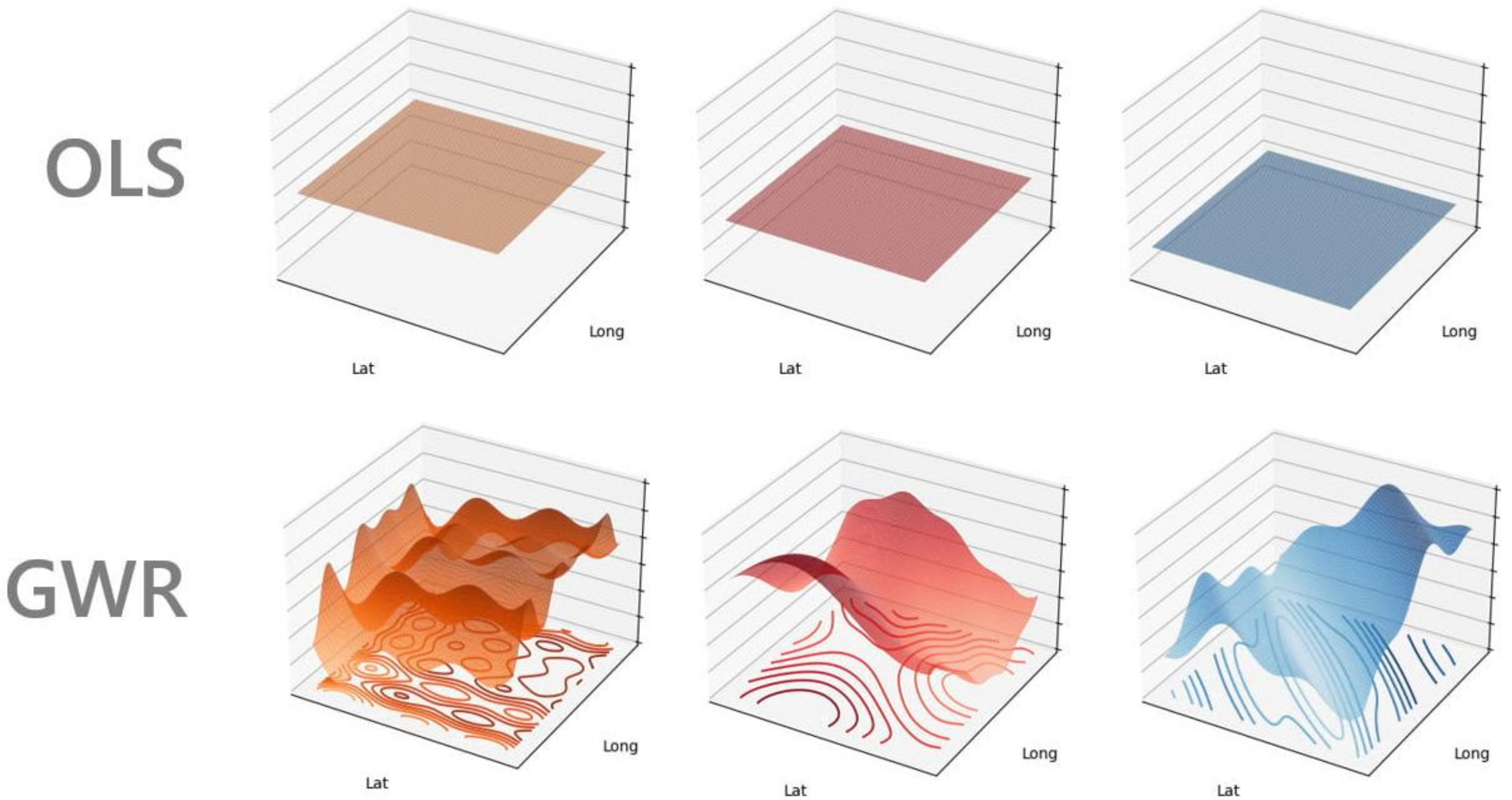
*(u_i, v_i) is the location of observation i

$$y_i = \beta_0(u_i, v_i) + \sum_k \beta_k(u_i, v_i)x_{ik} + \varepsilon_i$$

*(u_i, v_i) is the location of observation i

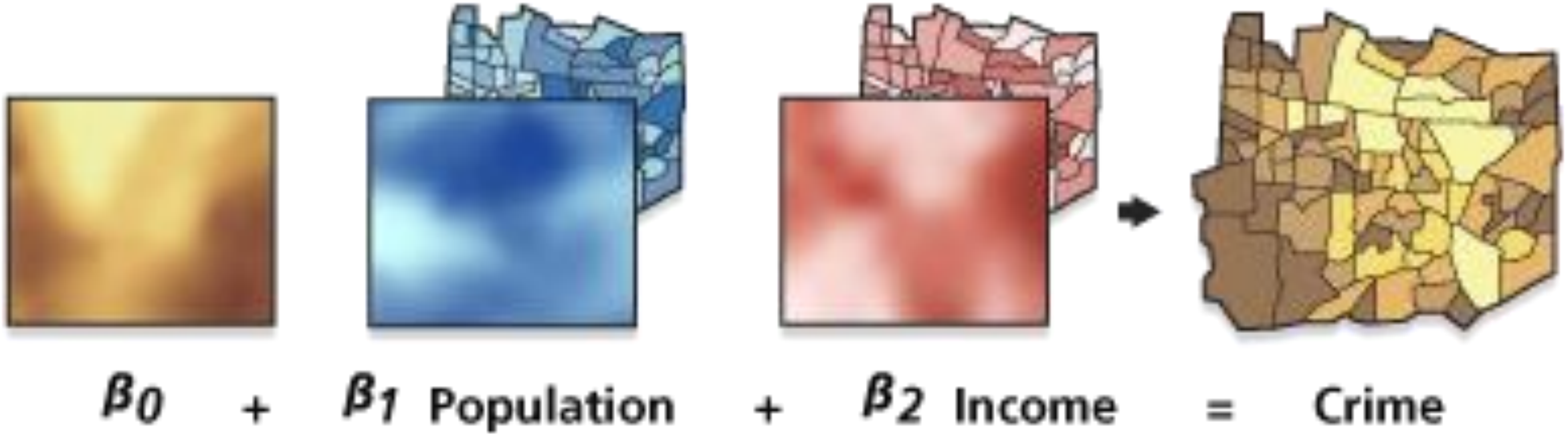
Geographically weighted regression

Coefficients can vary!



From: https://www.esri.com/arcgis-blog/products/arcgis-pro/analytics/discovering-spatial-relationships-with-multiscale-geographically-weighted-regression/

Geographically weighted regression

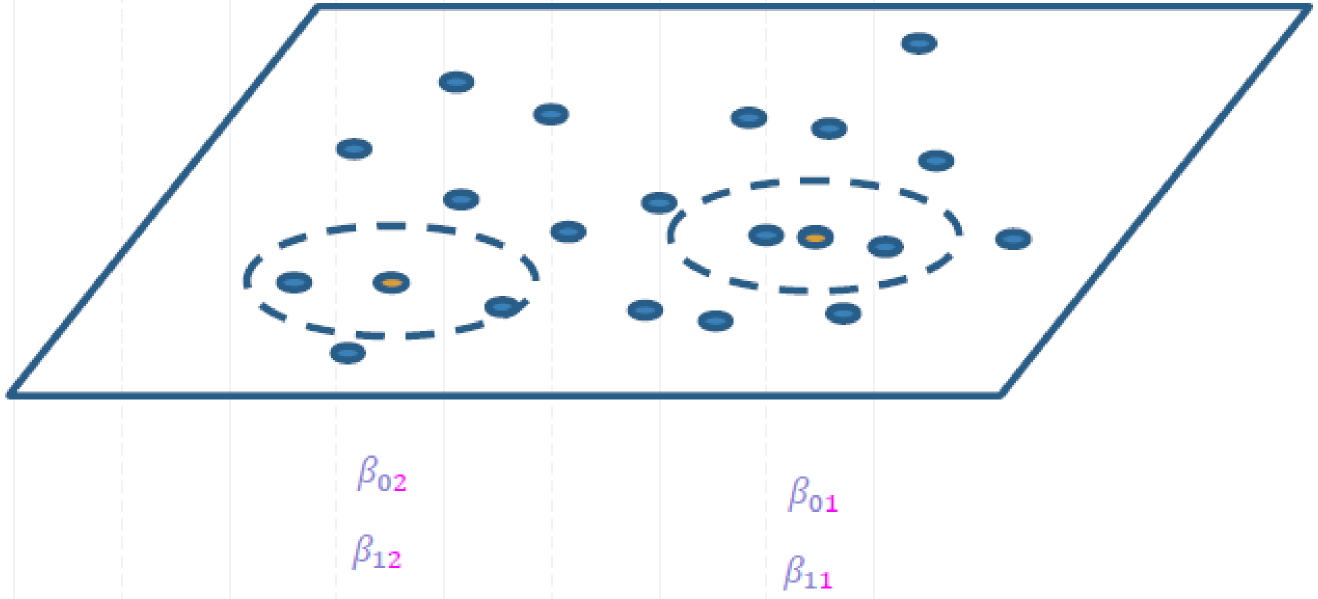


From https://pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-statistics/geographically-weighted-regression.htm

- **Local Regression**

$$y_i = \beta_{0i} + \beta_{1i} x_{i1} + \varepsilon_i$$

Varying over i



Local Regression (Kernel Function)

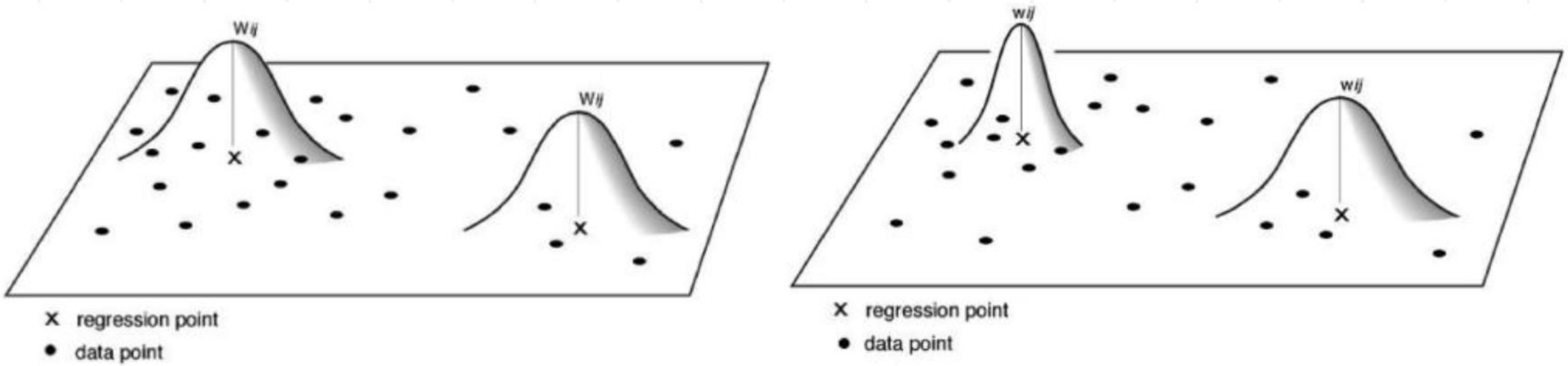


Figure 4. Reproduced from [2]. Examples of fixed (**left**) and adaptive (**right**) bandwidth kernels.

Source: Oshan, T. M., Li, Z., Kang, W., Wolf, L. J., & Fotheringham, A. S. (2019). MGWR: A Python implementation of multiscale geographically weighted regression for investigating process spatial heterogeneity and scale. ISPRS International Journal of Geo-Information, 8(6), 269.

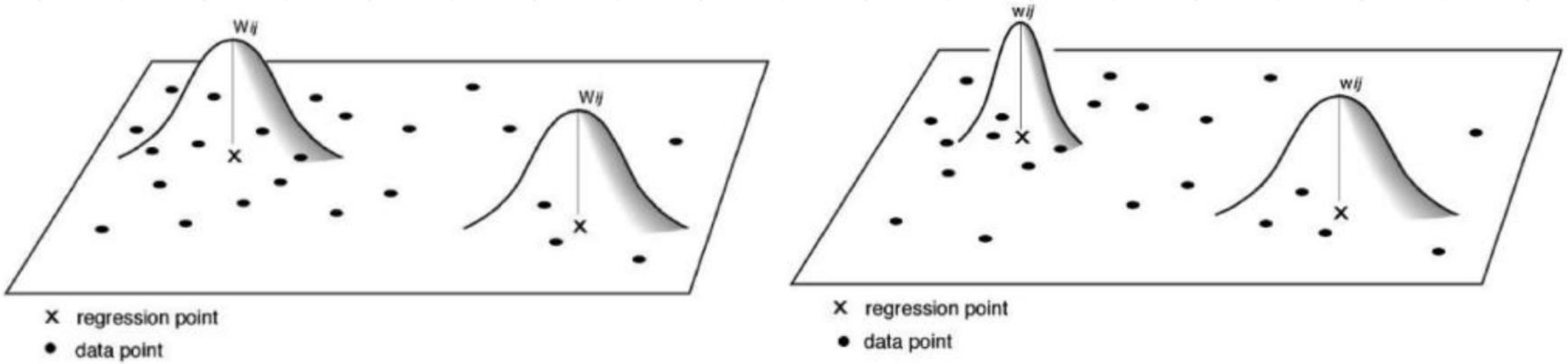
Kernel Function

Kernel Function	Form
Fixed Gaussian	$w_{ij} = exp(-d_{ij}^2/\theta^2)$
Fixed bi-square	$w_{ij} = \begin{cases} (1 - \frac{d_{ij}^2}{\theta^2})^2 & d_{ij} < \theta \\ 0 & d_{ij} > \theta \end{cases}$
Adaptive bi-square	$w_{ij} = \begin{cases} (1 - \frac{d_{ij}^2}{\theta_{i(k)}^2})^2 & d_{ij} < \theta_{i(k)} \\ 0 & d_{ij} > \theta_{i(k)} \end{cases}$
Adaptive Gaussian	$w_{ij} = exp(-d_{ij}^2/\theta_{i(k)}^2)$

GWR Practical Issues

- Choice of bandwidth
- Parameter inference

Local Regression (Kernel Function)



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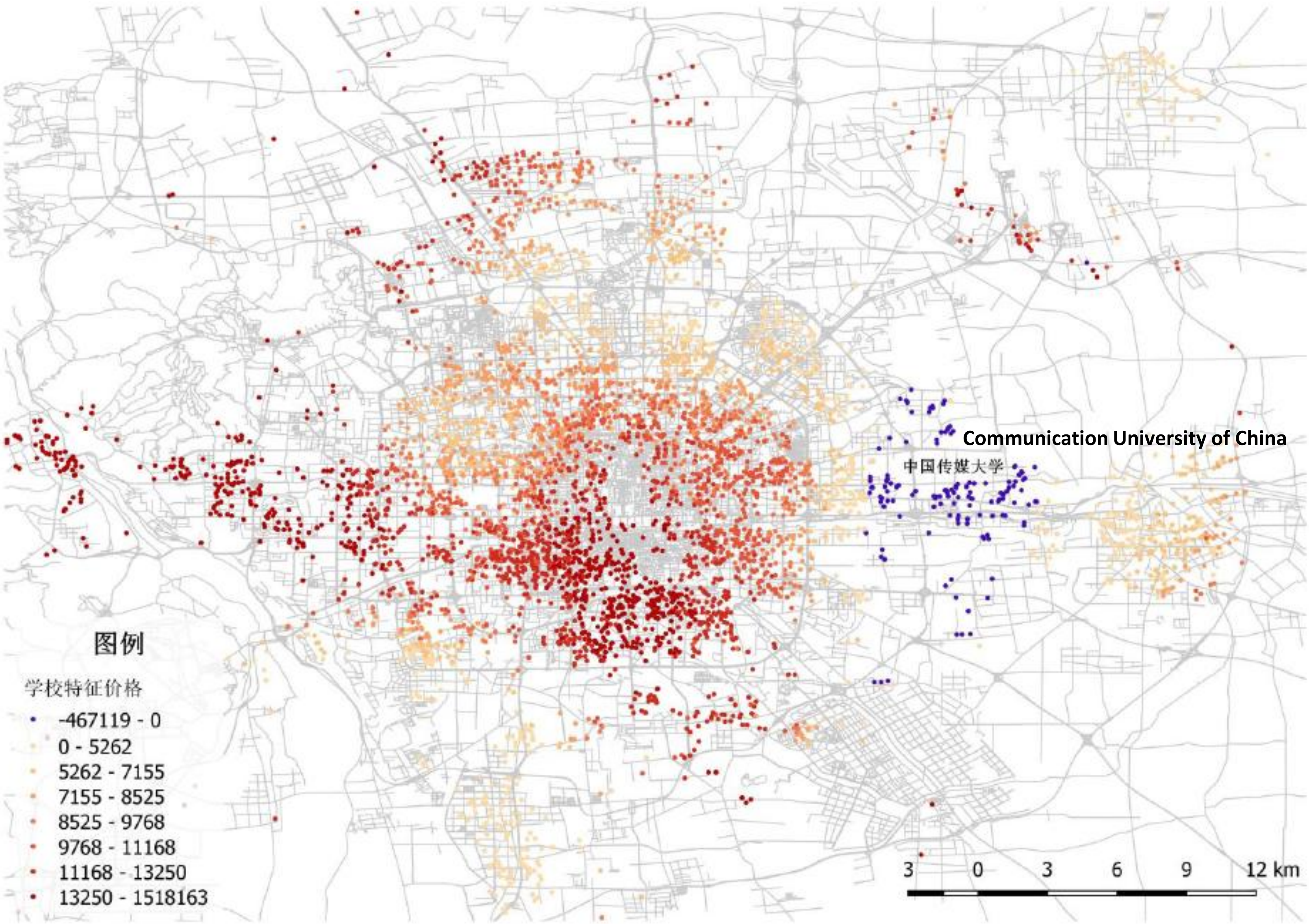
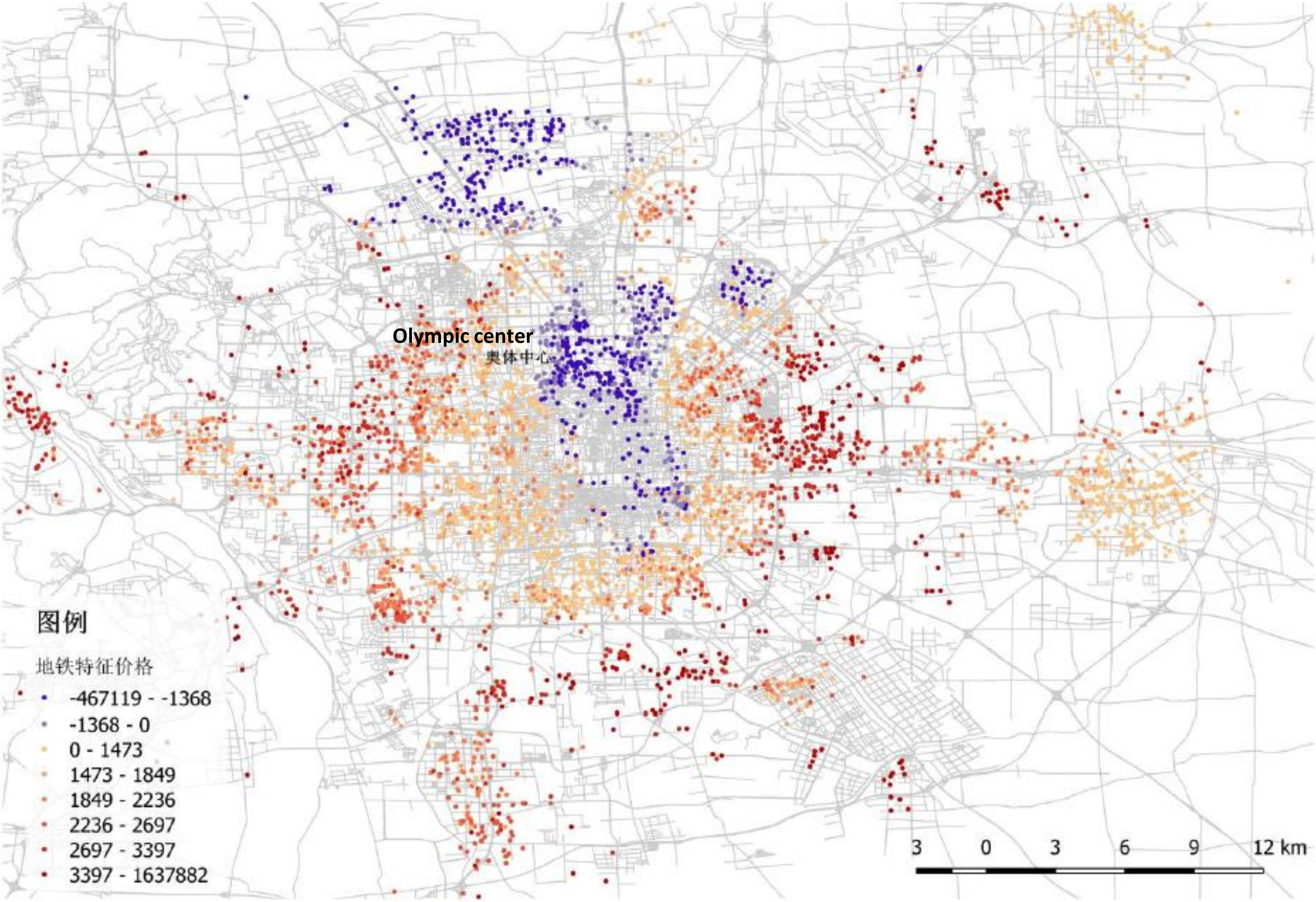
GWR Output Example 1

Hedonic Price of School Quality in Beijing



GWR Output Example 2

Hedonic Price of Distance to Nearest Subway in Beijing



Note:

- OLS regression: can be implemented by either ***statsmodels*** or ***Pysal*** packages.
- Spatial regressions can only be performed using Pysal

Feature	**statsmodels**	**PySAL**
Main Use	General statistical and econometric models	Spatial data analysis and modeling
Supports Spatial Models	No	Yes (e.g., spatial lag, error, GWR)
Time Series Support	Yes (ARIMA, SARIMAX, etc.)	No
Input Data Type	Tabular (DataFrame)	Spatial (GeoDataFrame, shapefile)
Diagnostics	Classical (e.g., R^2, F-test, hetero tests)	Spatial (e.g., Moran's I, LISA)

Reference and Further Reading

Software

- GWR4 (https://sgsup.asu.edu/sparc/gwr4)
- MGWR (https://sgsup.asu.edu/sparc/mgwr)

- Fotheringham, A. S., Brunsdon, C., & Charlton, M. (2003).*Geographically weighted regression: the analysis of spatially varying relationships*. John Wiley & Sons.

- Brunsdon, C., Fotheringham, A. S., & Charlton, M. E. (1996). Geographically weighted regression: a method for exploring spatial nonstationarity. *Geographical analysis*,*28*(4), 281-298.